



Lab Mid - B

CSE 4108

Structured Programming I Lab

Submit individual C files for each of your tasks in the Google Classroom Assignment. The files should be named in the format: "<ID>_Task<TaskNumber>.c". For instance, if your ID is 123456, for task 3, the file name should be "123456_Task3.c". Do not submit executable files.

Tasks

Task 1: Write a C program that functions as a simple calculator, supporting the following operations: addition, subtraction, multiplication, division, modulus, exponentiation, and tetration. There will be 3 inputs, the first two will be the two numbers n_1 and n_2 , and the third input will be the *operator* which may be one of the following characters: '+' (addition), '-' (subtraction), '*' (multiplication), '/' (integer division), '%' (integer modulus), '^' (exponentiation), and '#' (tetration). The output will be a single integer, the result of $n_1 \text{ operator } n_2$. If the result is too large even for unsigned long long int, then print OVERFLOW.

Note: In case you are unfamiliar with the tetration operation, it works like "repeated exponentiation". For example, 2 tetrated to 4 would be $2^{2^{2^2}}$. And exponentiation is right associative so it is performed as $2^{(2^{(2^2)})} = 2^{(2^4)} = 2^{16} = 65536$

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Sample Input	Sample Output
4 3 ^	64
4 3 #	OVERFLOW
4 2 #	256

Task 2: Suppose your friend came up with the following hypothesis: "There exists such a positive integer n that for each positive integer m , the number $n \times m + 1$ is a prime number". Your friend asks you to help test his hypothesis. For now, you decided to test whether the hypothesis works for any integer from 0 to 100000. Your friend will give you any number within this range, and you will try to find such a positive integer m for which $n \times m + 1$ is not a prime

$2 \ 4 \# \rightarrow 2^{2^{2^2}}$

②

number. If such a number cannot be found, then that means his hypothesis is correct.

Write a C program that will try to find counterexamples to disprove the hypothesis. The input will be a single integer n , and the output will be the smallest value for which $n*m+1$ is not a prime number.

Sample Input	Sample Output
3	1
4	2
30	3

Task 3: In the SCRABBLE crossword game, players form words using small tiles each containing a letter with a face value. The face value varies from one letter to another, based on the letter's rarity. The letters which are more common in the English language are of less value, and uncommon letters have more value. Multiple letters can have the same face value. The face values of the letters are given as follows:

Letters	Face Value
AEILNORSTU	1 $A=1, E=1, I=1, \dots$
DG	2
BCMP	3
FHVWY	4
K	5
JX	8
QZ	10 $Q=10, Z=10$

Write a C program that computes the total value of a word by summing the face values of its letters. The input will be a single word containing either uppercase or lowercase alphabets followed by a newline. The output will be the total value of the word.

Sample Input:



3

Sample Output:

12

Task 4: Write a C program that reads the marks of a fixed number of students and performs some basic analysis. The first input will be the number of students n . Next n lines will contain the marks of each student, where $0 \leq \text{marks} \leq 100$. The program should display the number of students passed (considering 40% as the pass mark), number of students failed, highest mark, lowest mark, and average mark.

Sample Input	Sample Output
Enter number of students: 5 Enter mark of student 1: 65 Enter mark of student 2: 48 Enter mark of student 3: 72 Enter mark of student 4: 90 Enter mark of student 5: 35	Passed: 4 Failed: 1 Highest Mark: 90 Lowest Mark: 35 Average Mark: 62.00